Signed Filtering before Majorization: Exact Literal Reconstruction, a Canonical Signed Energy, and Sharp Separation from Positive Resonance

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Abstract

The positive TPC resonance route takes absolute values before the low-window reconstruction filter. This is safe, but TPC-106 shows that distinct affine maps can remain perfectly aligned. We return to the exact TPC-93 expansion and move the filter ahead of majorization.

For the completely opened, frequency-free literal atoms p, let a_p retain the source coefficient, both Möbius signs, the extracted $\mu(B)$, all masks and weights, and the one physical normalization. Let $\psi_p(r)$ be the exact unit phase, ω_p the native multiplier, and H_p the inherited resonance width. With the signed reconstruction multiplier $\mathfrak{m}_X(r)$, the resonant contribution is exactly

$$\mathcal{S}_{\mathrm{res},X} = \sum_r \mathfrak{m}_X(r) \sum_p a_p \psi_p(r) \mathbf{1}_{E_{H_p}}(r\omega_p).$$

Finite reordering gives the filter-first identity

$$\mathcal{S}_{\mathrm{res},X} = \sum_p a_p \mathcal{K}_p, \qquad \mathcal{K}_p = \sum_r \mathfrak{m}_X(r)\psi_p(r)\mathbf{1}_{E_{H_p}}(r\omega_p).$$

Consequently the filtered-atom majorant $\sum_p |a_p||\mathcal{K}_p|$ is always no larger than the pre-filter positive carrier.

We also identify the canonical unfiltered signed frequency energy

$$\mathcal{E}_{\mathrm{sig},X} = \sum_r \left| \sum_p a_p \psi_p(r) \mathbf{1}_{E_{H_p}}(r\omega_p) \right|^2$$

and prove its exact positive-semidefinite TT^* Gram formula. Thus

$$|\mathcal{S}_{\mathrm{res},X}| \le \|\mathfrak{m}_X\|_2 \mathcal{E}_{\mathrm{sig},X}^{1/2}.$$

Two sharp examples delimit the result. Signed atoms can cancel completely while the positive carrier is nonzero; conversely, a phase gauge can saturate the positive carrier. Hence filtering before majorization preserves a real cancellation channel but does not manufacture cancellation. The reconstruction and inequalities are L0, and their imported literal dictionary is L1. The growing fixed-h_0 bound for $\mathcal{E}_{\mathrm{sig},X}$ is a new L2 arithmetic gate and is not proved.

Keywords: fixed-shift Möbius correlation; signed reconstruction filter; resonance energy; TT^*; positive majorant; parity barrier.

Audit convention. L0 denotes finite identities and sharp finite examples, L1 their exact attachment to the literal fixed-h_0 dictionary, and L2 only a growing fixed-shift arithmetic estimate. No cancellation created by a model coefficient is reported as progress on the physical packet.

1 Why the order of operations matters

TPC-93 exports the complete low window into fixed-determinant affine fibers while retaining the two Möbius signs, content extraction, both polarizations, exact phases, and provenance [4]. TPC-99–TPC-102 then construct a positive resonance carrier by taking absolute values and derive exact finite-group bounds for it [1, 5]. This route is sufficient, not equivalent to the signed low window.

TPC-105 removes identical-map duplication, while TPC-106 proves that pairwise distinct maps can still align perfectly [2, 3]. The prescribed fallback in TPC-MVP1 is therefore

$$\text{signed filter} \longrightarrow \text{arithmetic estimate} \longrightarrow \text{majorization},$$

with no early deletion of phases or Möbius signs [6]. The purpose of this paper is to make that fallback exact.

2 The frequency-free literal atom dictionary

Let $q = q_X$ and $G = \mathbb{F}_q^\times$. Extend every frequency weight by zero outside the actual low window. Let $\mathcal{P}_X$ be the finite set of fully opened, frequency-free atoms in the unresolved q_X-invertible nonconstant carrier with inherited branch length $1 \le N_p \le q_X$. Branches with $N_p > q_X$ have no resonant nonzero frequency and remain in the separate generic return. The conductor-one and $q_X \mid u$ pieces may be absent only because their disjoint exact returns were already recorded in TPC-98 and TPC-100 [1]. An atom $p \in \mathcal{P}_X$ retains:

- the source row, polarization, opposite row, interval component, content c, projector κ, extracted factor B, and resolved point z;
- the two reduced affine values

$$D_p = d_p + s_p z, \qquad V_p = u_p + a_p' z, \qquad s_p u_p - a_p' d_p = h_0 \neq 0;$$

- the factors $\mu(\kappa)\mu(B)\mu(D_p)\mu(V_p)$, $\mathbf{1}_{(B,V_p)=1}$, every inherited mask, and the exact physical weight;
- the native multiplier $\omega_p \in G$, inherited width H_p, and every r-dependent unit phase.

To avoid a collision of notation, write $\alpha_p \in \mathbb{C}$ for the complete r-independent signed coefficient of the atom, including one copy of the global normalization. Write $\psi_p : G \to \mathbb{C}$ for its exact phase, with $|\psi_p(r)| \in \{0, 1\}$; a zero records an inactive source condition. In the TPC-93 notation, the additive part of this phase contains

$$e_{cq_X}(-r\varepsilon_\theta \Omega_p z),$$

and the remaining r-dependent source phase is retained rather than set to one.

Equivalently, the allowed atom set and factor split may be recorded as

$$\mathcal{P}_X = \left\{ (\theta, c, \kappa, z) : \begin{array}{l} z \in I^*_{\theta,c,\kappa},\ 1 \le N_{\theta,c,\kappa} \le q_X,\ q_X \nmid u_p, \\ \omega_p \in G,\ \text{and every inherited literal mask is active} \end{array} \right\},$$

and

$$\begin{aligned} \alpha_p &= \mathcal{N}_X c_{\theta,X} \mu(\kappa_p)\mu(B_p)\mu(D_p)\mu(V_p)\mathbf{1}_{(B_p,V_p)=1}\mathcal{W}_p^{(0)}, \\ \psi_p(r) &= \zeta_{p,r} e_{c_p q_X}(-r\varepsilon_\theta \Omega_p z)\mathcal{W}_p^{(r)}. \end{aligned}$$

Here $\mathcal{N}_X$ is the single inherited global normalization, $\mathcal{W}_p^{(0)}$ is the product of all remaining r-independent literal weights and masks, and $\mathcal{W}_p^{(r)}$ is the product of the remaining r-dependent unit phases or zero source masks. This is a partition of the TPC-93 master coefficient, not a new renormalization.

Let

$$\rho_p(r) = \mathbf{1}_{E_{H_p}}(r\omega_p), \qquad \mathfrak{m}_X(r) = \mathfrak{m}_{K,X}(r)\mathbf{1}_{\{0 < d_{q_X}(r,0) \le R_X\}}.$$

The filter is signed and complex; the previously used positive weight is $|\mathfrak{m}_X(r)|$.

Theorem 2.1 (Exact signed resonant reconstruction). *Under the TPC-93 source-to-child inverse and exact-content crosswalk, the complete signed resonant contribution is*

$$\boxed{\mathcal{S}_{\mathrm{res},X} = \sum_{r\in G} \mathfrak{m}_X(r) \sum_{p\in\mathcal{P}_X} \alpha_p \psi_p(r)\rho_p(r).} \tag{1}$$

Every literal opened atom occurs once. No polarization, content factor, phase, mask, or fixed-h_0 determinant is changed.

Proof. Start with the finite TPC-93 sector master formula, split the complete outer coefficient into the signed reconstruction multiplier and the remaining source phase, and open the resolved fiber point z. The source-to-child inverse makes (θ, c, κ, z) a frequency-free atom, while the resonance test supplies $\rho_p(r)$. Reindexing finite sums gives (1). The exact content extraction is the step that retains determinant h_0 and the factor $\mu(B)\mathbf{1}_{(B,V_p)=1}$. □

Remark 2.2 (Scope). Theorem 2.1 imports the previously proved literal dictionary. It is not a new estimate for the inner Möbius sum.

3 Filter first, majorize second

Definition 3.1 (Atomwise filtered kernel). For $p \in \mathcal{P}_X$, put

$$\mathcal{K}_p = \sum_{r\in G} \mathfrak{m}_X(r)\psi_p(r)\rho_p(r). \tag{2}$$

Theorem 3.2 (Filter-before-majorant identity and domination). *One has*

$$\mathcal{S}_{\mathrm{res},X} = \sum_{p\in\mathcal{P}_X} \alpha_p \mathcal{K}_p, \tag{3}$$

$$|\mathcal{S}_{\mathrm{res},X}| \le \underbrace{\sum_p |\alpha_p||\mathcal{K}_p|}_{\mathcal{C}_{\mathrm{filt},X}} \le \underbrace{\sum_p |\alpha_p| \sum_r |\mathfrak{m}_X(r)|\rho_p(r)}_{\mathcal{C}_{\mathrm{preabs},X}}. \tag{4}$$

Proof. Interchange the two finite sums in (1). The first inequality is the triangle inequality after filtering. For the second, apply the triangle inequality inside (2) and use $|\psi_p(r)| \le 1$. □

Remark 3.3 (Relation to the positive census). After the literal fiber mass is opened into the atoms p, the last term of (4) is the corresponding pre-filter positive carrier. If one first replaces the signed fiber by a uniform absolute mass, it may be still larger.

Proposition 3.4 (The separation can be complete). *There exist two signed atoms with identical (ψ, ρ) and coefficients $\alpha_1 = 1$, $\alpha_2 = -1$ for which*

$$\mathcal{S}_{\mathrm{res}} = 0, \qquad \mathcal{C}_{\mathrm{filt}} = 2|\mathcal{K}_1|, \qquad \mathcal{C}_{\mathrm{preabs}} = 2\sum_r |\mathfrak{m}(r)|\rho_1(r).$$

If $\mathcal{K}_1 \ne 0$, both positive quantities are nonzero.

Proof. The two atom contributions cancel at every frequency before the filter is applied. □

Proposition 3.5 (No automatic phase saving). *Given any nonzero filter $\mathfrak{m}$ and nonempty resonance support ρ, there is a unit phase ψ on that support such that*

$$\left|\sum_r \mathfrak{m}(r)\psi(r)\rho(r)\right| = \sum_r |\mathfrak{m}(r)|\rho(r).$$

Proof. On points where $\mathfrak{m}(r) \ne 0$, choose $\psi(r) = \overline{\mathfrak{m}(r)}/|\mathfrak{m}(r)|$; choose any unit value elsewhere. □

The last proposition is a coefficient-blind sharpness statement. The literal phases are not freely chosen, but no theorem may infer a saving merely from their unit modulus.

4 The canonical signed energy and exact TT^*

Define the unfiltered signed frequency profile

$$F_X(r) = \sum_{p\in\mathcal{P}_X} \alpha_p\psi_p(r)\rho_p(r) \tag{5}$$

and its energy

$$\mathcal{E}_{\mathrm{sig},X} = \sum_{r\in G} |F_X(r)|^2. \tag{6}$$

This is canonical for the full finite frequency group, not an extremal minimum: restricting the sum to $\operatorname{supp}\mathfrak{m}_X$ can only decrease it and gives another valid sufficient bound.

Theorem 4.1 (Exact signed Gram identity)**.** *Let*

$$\Gamma_{p,p'} = \sum_{r\in G} \psi_p(r)\overline{\psi_{p'}(r)}\rho_p(r)\rho_{p'}(r). \tag{7}$$

Then Γ is positive semidefinite and

$$\boxed{\mathcal{E}_{\mathrm{sig},X} = \sum_{p,p'} \alpha_p\overline{\alpha_{p'}}\Gamma_{p,p'}.} \tag{8}$$

Moreover,

$$|\mathcal{S}_{\mathrm{res},X}| \le \|\mathfrak{m}_X\|_2\mathcal{E}_{\mathrm{sig},X}^{1/2}. \tag{9}$$

Proof. Expand (6) using (5) to obtain (8). The matrix is the Gram matrix of the vectors $r \mapsto \psi_p(r)\rho_p(r)$. Finally, $\mathcal{S}_{\mathrm{res},X} = \langle\overline{\mathfrak{m}_X}, F_X\rangle$ up to the chosen inner-product convention, so Cauchy–Schwarz gives (9). $\square$

Corollary 4.2 (A precise signed replacement gate)**.** *Since the inherited low-window filter satisfies $\|\mathfrak{m}_X\|_2 \ll_K 1$, the bound*

$$\mathcal{E}_{\mathrm{sig},X}^{1/2} \ll X^{o(1)}Q_X^2 \tag{10}$$

is sufficient to return the signed resonant contribution at the native TPC-102 scale.

Remark 4.3 (Why this remains arithmetic)*.* The off-diagonal entries of (8) retain products of the actual Möbius signs, source phases, and masks. Bounding (10) is therefore not finite spectral algebra. It is a growing fixed-h_0 arithmetic estimate.

5 What information survives

The signed profile in (5) preserves simultaneously:

(i) the signed reconstruction multiplier;

(ii) the r-dependent source and determinant phases;

(iii) both Möbius signs and the extracted $\mu(B)$;

(iv) the coprimality mask $\mathbf{1}_{(B,V)=1}$;

(v) the actual moving affine row and fixed determinant h_0;

(vi) both polarizations and all outer keys; and

(vii) the one physical post-aggregation normalization.

Replacing any item by an independent model variable changes the Gram matrix. Conversely, identical signed atoms may be combined before absolute values provided their complete provenance remains recoverable.

Proposition 5.1 (Positive failure does not imply signed failure)**.** *A polynomial obstruction in the principal or distinct-map term of a positive carrier does not yield a lower bound for $\mathcal{E}_{\mathrm{sig},X}$ or $|\mathcal{S}_{\mathrm{res},X}|$.*

Proof. Proposition 3.4 gives an exact finite example with a nonzero positive carrier and zero signed profile. Therefore no implication exists without additional literal information. $\square$

6 Deterministic regression certificate

The script `experiments/tpc107_certificate.py` checks:

- the frequency-first and atom-filtered forms of the signed sum;
- the positive-majorant chain;
- the exact Gram identity and positive-semidefinite quadratic form on deterministic rational/complex examples;
- complete signed cancellation with positive absolute mass; and
- phase-gauge saturation.

These are finite regression checks, not numerical evidence for (10).

7 Claim boundary and route verdict

Proved

The exact literal signed resonant sum can be filtered before majorization. The resulting atomwise majorant never exceeds the corresponding pre-filter positive carrier. The canonical unfiltered signed frequency energy has the positive-semidefinite TT^* identity (8), and (10) is a sufficient replacement gate. Sharp finite examples show both complete improvement and complete saturation. These are L0 statements; the imported source-to-atom crosswalk is L1.

Not proved

This paper does not prove:

- (10) for the growing physical coefficient;
- a Möbius or Liouville cancellation theorem on the actual fixed-h_0 carrier;
- that the literal phases realize either sharp finite example;
- closure of the generic affine sector, high-frequency return, full-block reassembly, or the strict physical endpoint; or
- a parity breakthrough, prime-pair lower bound, or twin-prime theorem.

Nothing is specialized to $h_0 = 2$.

Verdict

TPC-107 opens the signed fallback corridor exactly, but it does not walk through the parity-facing door. Its gain is architectural: cancellation discarded by the positive route is now represented by one explicit Gram energy. TPC-108 must determine what arithmetic input could control the analogous literal generic-affine Möbius sums without disguising an unproved Chowla-type estimate.

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